

16.1.2023

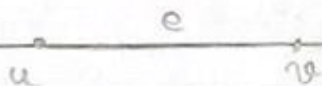
Unit - IV

Graph Theory

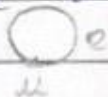
17.1.2023* Graph:

A graph $G = (V(G), E(G))$ consists of 2 finite sets where $V(G)$ is the vertex set of G which is a non-empty set of elements called vertices. It is denoted by V and $E(G)$ is the edge set of G which is possibly a non-empty set of elements called edges and is denoted by E . Such that each edge in G is assigned a pair of vertices called end-vertices.

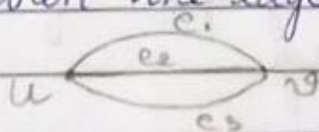
Note: 1) If e is an edge with end vertices u & v then e is said to join u and v .



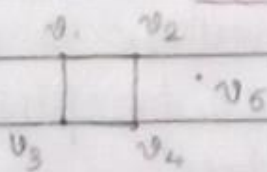
2) In a graph G with edge e and same end vertices is called a loop or Any vertex is joined itself by an edge is called a loop or self loop.



3) In a graph G , 2 or more edges having the same end vertices, then the edges are called parallel edges.



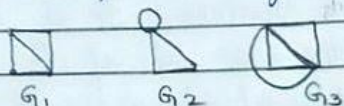
4) In a graph G , if any vertex which is not the end of any vertex is called an isolated vertex.



$v_5 \rightarrow$ isolated vertex

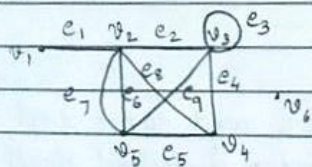
2) Adjacent Vertices - 2 vertices joined by an edge are called adjacent vertices. If 2 edges u and v are said to be adjacent if there exists an edge b/w them.

6) Simple graph - G is said to be a simple graph if it has no parallel edges and no self-loops.



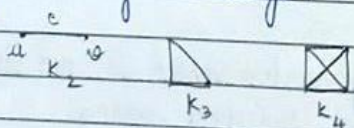
$G_1 \rightarrow$ is a simple graph; $G_2, G_3 \rightarrow$ not simple since
 $G_2 \rightarrow$ self loop
 $G_3 \rightarrow$ parallel edges

Eg. graph:



$e_7, e_6 \rightarrow$ || edges $v_6 \rightarrow$ isolated vertex
 $e_8 \rightarrow$ self loop

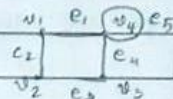
7) Complete graph - A simple graph G is said to be a complete graph if every pair of distinct vertices are joined by an edge.



* Complete graph of n vertices is denoted by K_n .
 * Max No. of edges in a complete graph of n vertices is $\frac{n(n-1)}{2}$ edges

8) An edge e of a graph G is said to be incident with a vertex v if v is an end vertex of e . In this case, we also say that v is incident with e .

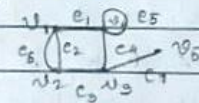
9) Adjacent Edges - 2 edges e_1 and e_2 which are incident with a common vertex are called adjacent edges.



* An edge is incident with either 1 or 2 vertices [1 $\rightarrow$ if it is self loop] whereas a vertex may be incident with finite no. of edges including zero.

10) Degree of Vertex - Let V be the vertex set of graph G . The degree of $v \in V$ is the no. of edges of G that are incident with v counting each loop twice i.e. it is the no. of times v is an end vertex of an edge.

Denoted by $d(v)$



$$d(v_1) = 3$$

$$d(v_2) = 3$$

$$d(v_3) = 3$$

$$d(v_4) = 4$$

$$d(v_5) = 1$$

$$\sum_{i=1}^5 d(v_i) = 3 + 3 + 3 + 4 + 1 = 14 = 2 \times 7 = 2e$$

* Theorem 1:

For a graph G with e edges and n vertices $v_1, v_2, \dots, v_n$ then $\sum_{i=1}^n d(v_i) = 2e$

In a graph G , each edge has 2 vertices so it contributes precisely 2 to sum of degrees of all

vertices i.e. when the degrees of all vertices are added, each edge is counted twice i.e.

$$\sum_{i=1}^n d(v_i) = 2e$$

* Odd vertices and Even Vertices

A vertex of a graph G is called odd or even depending on whether its degree is odd or even.

In the above eg,

$$\left. \begin{array}{l} d(v_1) = 3 = d(v_2) = d(v_3) \\ d(v_4) = 4 \end{array} \right\} \rightarrow \text{odd}$$

$$\left. \begin{array}{l} d(v_5) = 1 \\ d(v_6) = 4 \end{array} \right\} \rightarrow \text{even}$$

Note:- In a graph G , there are even no. of odd vertices

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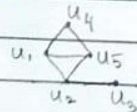
* Isomorphic Graph

A graph $G_1 = (V, E)$ is said to be isomorphic to $G_2 = (U, F)$ if there is 1 to 1 correspondence b/w the vertex set V & U and edge set E and F in such a way that $e_i \in E$ is an edge b/w v_i, v_j in G_1 , then the corresponding edge $f_i \in F$ in G_2 with u_i, u_j as end vertices. Such a pair of correspondence is called as graph isomorphism.

Eg: $G_1, G_2 \rightarrow$ are isomorphic



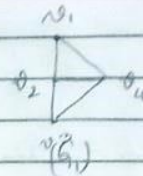
(G₁)



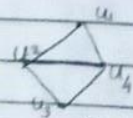
(G₂)

Note:- 1) Two graphs G_1 and G_2 are said to be isomorphic if they have same no. of edges and same no. of vertices

Eg: 2)



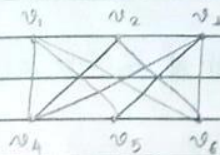
(G₁)



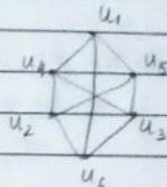
(G₂)

Here $G_1, G_2 \rightarrow$ contain 4 vertices and 5 edges and there is 1:1 correspondence b/w vertices and edges. $\therefore G_1, G_2 \rightarrow$ are isomorphic

Eg: 3)



(G₁)



(G₂)

Vertices = 6 Edges = 9

There is 1:1 correspondence b/w vertices and edges. $\therefore G_1, G_2$ are isomorphic

* Regular Graph

A graph G in which all the vertices are of equal degree is said to be a regular graph

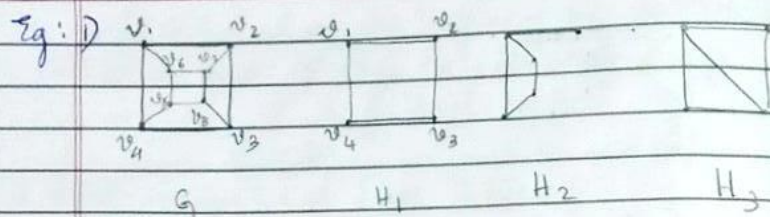
Eg. In eg: 3), G_1, G_2 are regular

* Sub Graphs

Let H be a graph with vertex set $V(H)$ and edge set $E(H)$ and G be another graph with vertex set $V(G)$ and edge set $E(G)$.

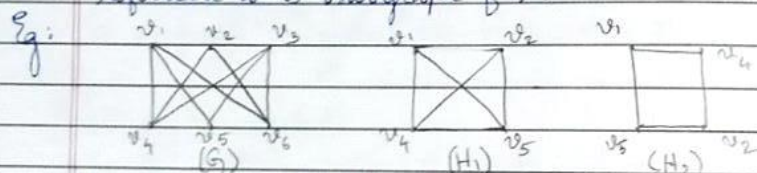
We say that H is a subgraph of G if $V(H)$ is contained in $V(G)$ i.e. $V(H) \subseteq V(G)$ or $E(H) \subseteq E(G)$.

and



In above example,
 H_1, H_2 are subgraphs of G
But H_3 is not a subgraph of G

Note:- Any graph isomorphic to subgraph of G is also referred to as subgraph of G .

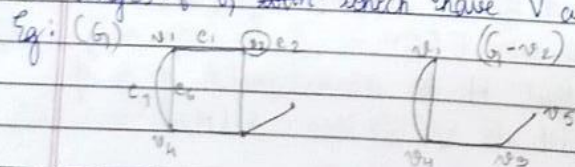


H_1 is a subgraph of G
 H_2 is isomorphic to H_1
 H_2 is also a subgraph of G

* Types:- In a graph G , which has atleast 2 vertices

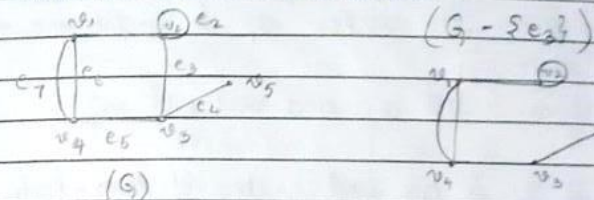
(i) Vertex deleted subgraph

In a graph G , which has atleast two vertices, then for any vertex $v \in G$, $(G-v)$ denotes subgraph of G with vertex set $(V-v)$ and whose edges are all those of G which are not incident with v i.e. $(G-v)$ is obtained from G by removing vertex v and all edges of G with which have v as an end vertex



(ii) Edge deleted subgraph

If $G=(V, E)$ is a graph and $e \in G$ be any edge, then $(G-e)$ is a subgraph of G having the same vertex set as G and each $(G-e)$ has its edge set i.e. $G-\{e\}$ is obtained from graph G by removing edge e but not its end vertices.

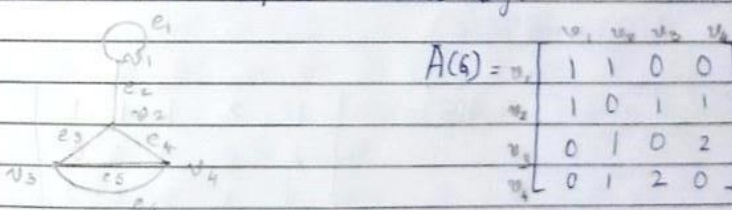


* Matrix Representation of Graphs

1) Adjacency matrix

Let G be a graph with n vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G wrt n vertices of G is a $n \times n$ matrix $A(G) = [a_{ij}]$, $(i, j = 1, 2, \dots, n)$ where $(i, j)^{th}$ entry a_{ij} is the no of edges joining the vertex v_i to vertex v_j .

Eg:



Note:- In the adjacency matrix $A(G)$, $a_{ii} = a_{ji}$ i.e., adjacency matrix is symmetric

1) If G has no loops then all the entries in the main diagonal are 0s

2) If there are no || edges, then all the entries of $A(G)$ are either 0s or 1s

2) Incidence Matrix

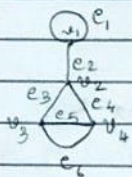
Let G be a graph with n vertices $v_1, v_2, \dots, v_n$ and m vertices edges $e_1, e_2, \dots, e_m$. The incidence matrix of G is $n \times m$ matrix given by

$$M[G] = [m_{ij}] \quad \begin{matrix} i=1, 2, \dots, n \\ j=1, 2, \dots, m \end{matrix}$$

$m_{ij} \rightarrow$ no. of times the vertex v_i is incidence on the edge e_j .

$$m_{ij} = \begin{cases} 0 & \text{if } v_i \text{ is not an end-vertex of } e_j \\ 1 & \text{if } v_i \text{ is an end-vertex of non-loop } e_j \\ 2 & \text{if } v_i \text{ is an end-vertex of loop } e_j \end{cases}$$

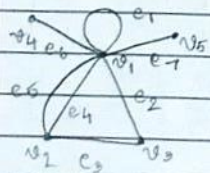
Eg:



	e_1	e_2	e_3	e_4	e_5	e_6
$M(G) = v_1$	2	1	0	0	0	0
v_2	0	1	1	1	0	0
v_3	0	0	1	0	1	1
v_4	0	0	0	1	1	1

* Problems:

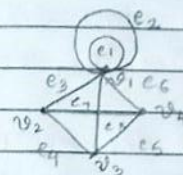
1) Find $A(G)$ and $M(G)$



	v_1	v_2	v_3	v_4	v_5
$A(G) = v_1$	1	2	1	1	1
v_2	2	0	1	0	0
v_3	1	1	0	0	0
v_4	1	0	0	0	0
v_5	1	0	0	0	0

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
$M(G) = v_1$	2	1	0	1	1	1	1
v_2	0	0	1	1	1	0	0
v_3	0	1	1	0	0	0	0
v_4	0	0	0	0	0	1	0
v_5	0	0	0	0	0	0	1

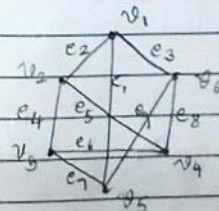
2) $A(G) = v_1 \begin{bmatrix} 2 & 1 & 1 & 1 \\ v_2 & 1 & 0 & 1 & 1 \\ v_3 & 1 & 1 & 0 & 1 \\ v_4 & 1 & 1 & 1 & 0 \end{bmatrix}$ Draw graph and find $M(G)$



	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
$M(G) = v_1$	2	2	1	0	0	1	1	0
v_2	0	0	1	1	0	0	0	1
v_3	0	0	0	1	1	0	1	0
v_4	0	0	0	0	1	1	0	1

3) Given $M(G)$ find graph. (Draw)

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8	e_9
$M(G) = v_1$	1	1	1	0	0	0	0	0	0
v_2	0	1	0	1	1	0	0	0	0
v_3	0	0	0	1	0	1	1	0	0
v_4	0	0	0	0	1	1	0	1	0
v_5	1	0	0	0	0	0	1	0	1
v_6	0	0	1	0	0	0	0	1	1



* Every path is a trail but every trail is not a path.

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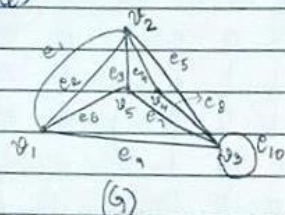
* Walk, Path, Trail, Cycles

* Walk - A walk in a graph is a finite sequence $w = v_0 e_1 v_1 e_2 v_2 \dots v_{k-1} e_k v_k$ whose terms are alternatively vertices and edges such that $1 \leq i \leq k$, the edge e_i has end vertices v_{i-1} and v_i .

We can also say that the above walk w is a walk from v_0 to v_k . The vertex v_0 is called the origin of the walk and vertex v_k is called terminus of the walk and the vertices $v_1, v_2, \dots, v_{k-1}$ are called the internal vertices of the walk.

In a walk w , the no. of edges is called the length of the walk (i.e. k is the length here)

Eg:



In the above graph G ,
 $w_1 = v_1 e_1 v_2 e_5 v_3 e_{10} v_4 e_8 v_5 e_2 v_6 e_9 v_7$

$w_2 = v_1 e_1 v_2 e_2 v_3 e_1 v_2$

$w_3 = v_1 e_6 v_5 e_3 v_2 e_4 v_4 e_8 v_3 v_9 v_1 = v_1 v_5 v_2 v_4 v_9 v_1$

Here w_1, w_2, w_3 are of lengths 5, 3, 5 respectively.

* Closed walk - In a graph G , if the origin and terminus are same, then it is closed walk otherwise it is an open walk.

Eg: $w_1, w_2 \rightarrow$ open walk (From eg in prev pg)
 $w_3 \rightarrow$ closed walk

* Trail - A walk w is said to be trail in which no edge is repeated.

Eg: In eg from prev pg, w_1 and w_2 are not trails because e_5 is repeated in w_1 & e_1 in w_2 . But w_3 is a trail

* Path - A walk is said to be a path if no vertex is repeated.

Eg: From (G) prev pg,

$w_4 = v_1 e_2 v_2 e_5 v_3 e_8 v_4$

w_4 is a path since no vertex is repeated. Also here no edge is repeated.

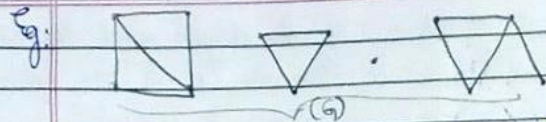
Note: Every path is a trail but vice versa not true.

Eg: w_3 is a trail but not a path since v_1 is

* Connected graph - A vertex u is said to be connected to a vertex v in a graph G if there is a path from u to v in G .

* Connected graph - A graph G is said to be connected if there exists a path b/w every pair of vertices. Otherwise graph G is disconnected.

* Component - If a graph G is disconnected then the no. of connected subgraph of G is called component. It is denoted by $W(G)$.



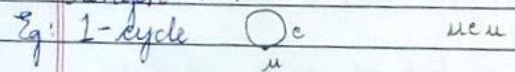
$W(G) = 4$ i.e. in G , no. of components is 4.

Note: If $W(G) = 1 \Leftrightarrow G$ is connected

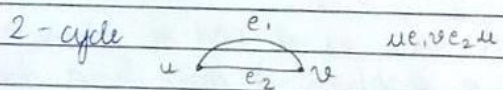
* Cycle - A non-trivial closed trail in a graph G is called a cycle if its origin and internal vertices are distinct. In other words, the closed trail $C = v_1 v_2 v_3 \dots v_m v_1$ is a cycle if C has atleast 1 edge and $v_1, v_2, v_3, \dots, v_m$ are all distinct.

The no. of edges in a cycle is called the length of the cycle.

In general, K -cycle means cycle of length K .



A 1-cycle is a self-loop.

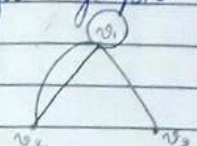


A 2-cycle has 2 edges $\rightarrow$ 11ed edge

3-cycle

A 3-cycle is often called a triangle

eg: Consider the foll graph



Adjacency matrix $A(G)$ is

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

length

$$\text{Now } A^2 = A \times A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = [b_{ij}] \end{matrix}$$

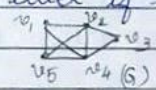
Here b_{ij} gives the no. of diff walks of length 2 from v_i to v_j (6 diff walks b/w v_1 & v_1 of length 2)

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* Euler trail

A trail in a graph G is called an Euler trail if it includes every edge of G .

$\therefore$ Trail is said to be Euler if every edge of G is in the trail exactly once.



* Tour - A closed walk of G which includes every edge of G exactly once.

* Euler Tour - An Euler tour of graph G which includes every edge of G exactly once. $\therefore$ Euler tour is just a closed Euler trail.

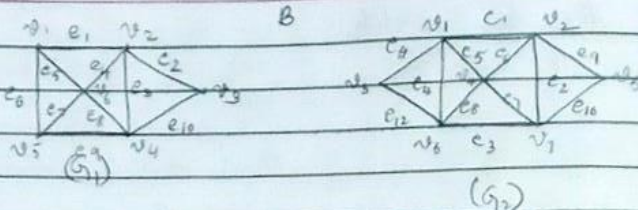
* Euler graph - A graph G is said to be an Euler graph if it has an Euler tour.

land-vertices
Bridges-edges

Konigsberg Bridge Problem $\rightarrow$ Not a Euler graph



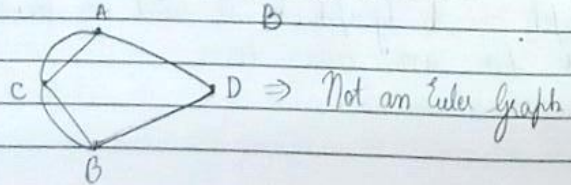
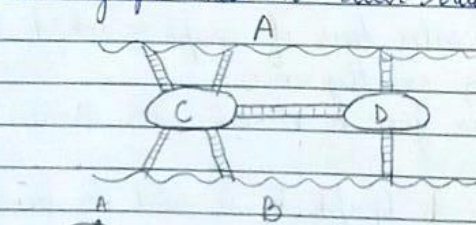
Halt at A/B/C/D - go through 7 bridges once and reach the same at end point



$T = v_1, v_4, v_6, v_7, v_4, v_2, v_7, v_5, v_2, v_1, v_4, v_3, v_1$ is an Euler tour for G_2

* Konigsberg Bridge Problem:

Early in the 18th century, the Swiss Mathematician Euler considered the problem of 7 Konigsberg Bridges. 2 islands C and D are formed by a river in the Prussian city of Konigsberg were connected to each other to the banks A and B with 7 Bridges as shown Below. The problem was to start at any one of the 4 land areas A, B, C, D and walk over each of the 7 bridges exactly once and return to starting point without swimming across river. Euler represented this by means of graph as shown below where vertices are land areas and edges are bridges. It is not possible to walk over each of the 7 bridges exactly once and return to starting point because graph has no Euler trail.



classmate

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Note: A connected graph G is Euler if and only if every vertex is even.

* Hamiltonian path

It is a path of graph G which contains every vertex of G .
+ Since no vertex is repeated in path, Hamiltonian path in G contains every vertex of G exactly once.

* Hamiltonian Cycle / Circle

A Hamiltonian Cycle in graph G is a cycle which contains every vertex of G .
+ Since no vertex is repeated apart from origin and terminus, a Hamiltonian Cycle with initial vertex v contains every other vertex of G exactly once and then ends at the vertex v .

* Hamiltonian Graph

A graph G is called Hamiltonian graph if it has a Hamiltonian cycle.

Note: 1) By simply deleting last edge of an Hamiltonian cycle, we get an Hamiltonian path.
2) A Non-Hamiltonian graph may possess a Hamiltonian path.

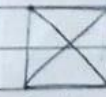
Eg:



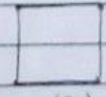
(G_1)



(G_2)



(G_3)



(G_4)

G_1 has no Hamiltonian path
 G_2, G_3 have Hamiltonian path
 G_4 has Hamiltonian cycle

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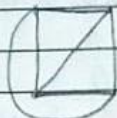
* Plane graphs and Planar graphs

A plane graph is a graph drawn on the plane in such a way that any pair of edges meet only at their end vertices if they meet at all.

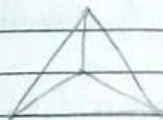
A planar graph is a graph which is isomorphic to plane graph i.e. can be re-drawn as plane graph.



(G₁)



(G₂)



(G₃)

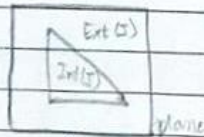
Here, all graphs are planar but (G₁) is not a plane graph.

Also (G₂) and (G₃) are re-drawings of (G₁).

* Jordan Curve

A Jordan Curve in a plane is a continuous non-self intersecting curve whose origin and terminus coincide.

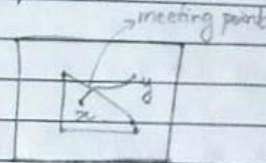
Note:- Any Jordan Curve divides the plane into 2 regions called the interior region and exterior region.



2) If J is the Jordan Curve in the plane, then the part of plane enclosed by J is called Interior of J [Int(J)]. The part of plane lying outside J is called Exterior of J [Ext(J)].

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3) If J is a Jordan curve and if x is a pt in Int(J) and y is a pt in Ext(J), then any curve or line joining x and y must meet J i.e. Jordan Curve at some point.



* Faces of a graph

A plane graph G partitions the plane into no. of regions called faces of the graph G .

No. of faces of G is denoted by f or $f(G)$.

Note:- 1) Any plane graph has exactly 1 exterior face.
2) Any other face bounded by a closed walk in the graph is called an interior face.



(G)

This graph has 4 faces
3 → Interior, 1 → Exterior

* Euler's formula

Let G be a connected plane graph and let m, e, f denotes no. of vertices, edges and faces of G respectively, then $m - e + f = 2$.

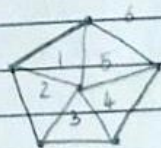
Eg: Above eg, $f = 4$
 $m = 4$
 $e = 6$

$$4 - 6 + 4 = 2$$

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Eg:



$$\begin{aligned} f &= 6 \\ n &= 6 \\ e &= 10 \\ 6 - 10 + 6 &= 2 \end{aligned}$$

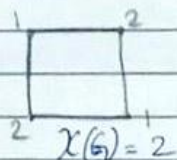
* Graph Colouring (Vertex Colouring)

→ must be simple graph (no self-loops or 1st edge)

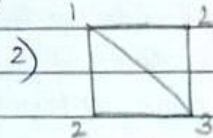
Let G be a graph. In the graph G we colour the vertices with different colours in such way that 2 adjacent vertices can't have the same colour.

* Chromatic No - The min. no. of colours that are used to colour a graph is called Chromatic number. Denoted by $\chi(G)$

Eg 1)

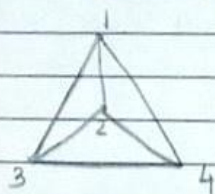


$$\chi(G) = 2$$



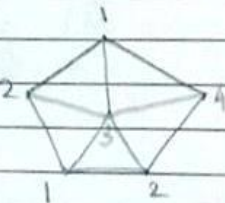
$$\chi(G) = 3$$

3)



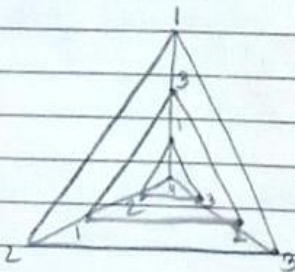
$$\chi(G) = 4$$

4)



$$\chi(G) = 5$$

5)



$$\chi(G) = 4$$

29.1.2023

Unit - V Group Theory

* Binary Operation

Let A be any non-empty set. A mapping defined from $A \times A$ to A is called binary composition on A .

$$A \times A \xrightarrow{f} A$$

They are denoted by $*, \circ, \cdot, \times, +$

$$\text{i.e. } a, b \in A \Rightarrow a * b \in A$$

then $*$ is a binary composition / operation on A .

* Group

Let G be any non-empty set and $*$ be a binary composition on G . The algebraic structure $(G, *)$ is said to be a group if it satisfies the following axioms:-

G_1 : Closure axiom - Let $a, b \in G$, then $a * b \in G$

G_2 : Associative axiom - If $a, b, c \in G$, then $a * (b * c) = (a * b) * c$

G_3 : Identity axiom - For any $a \in G$, there exists a unique element $e \in G$ such that $a * e = e * a = a$

Here e is called identity element of G .

G_4 : Inverse axiom - For each $a \in G$, $\exists$ a unique element $b \in G$ such that

$a * b = b * a = e$ where $e \in G$ is identity element. Here b is called inverse of element a or vice versa.